Habitual intake of flavonoid subclasses and incident hypertension in adults

Background: Dietary flavonoids have beneficial effects on blood pressure in intervention settings, but there is limited information on habitual intake and risk of hypertension in population-based studies. Objective: We examined the association between habitual flavonoid intake and incident hypertension in a prospective study in men and women. Design: A total of 87,242 women from the Nurses' Health Study (NHS) II, 46,672 women from the NHS I, and 23,043 men from the Health Professionals Follow-Up Study (HPFS) participated in the study. Total flavonoid and subclass intakes were calculated from semiquantitative food-frequency questionnaires collected every 4 y by using an updated and extended US Department of Agriculture database. Results: During 14 y of follow-up, 29,018 cases of hypertension in women and 5629 cases of hypertension in men were reported. In pooled multivariate-adjusted analyses, participants in the highest quintile of anthocyanin intake (predominantly from blueberries and strawberries) had an 8% reduction in risk of hypertension [relative risk (RR): 0.92; 95% CI: 0.86, 0.98; $P < 0.03$] compared with that for participants in the lowest quintile of anthocyanin intake; the risk reduction was 12% (RR: 0.88; 95% CI: 0.84, 0.93; $P < 0.001$) in participants ≤60 y of age and 0.96 (0.91, 1.02) in participants >60 y of age (P for age interaction = 0.02). Although intakes of other subclasses were not associated with hypertension, pooled analyses for individual compounds suggested a 5% (95% CI: 0.91, 0.99; $P = 0.005$) reduction in risk for the highest compared with the lowest quintiles of intake of the flavone apigenin. In participants ≤60 y of age, a 6% (95% CI: 0.88, 0.97; $P = 0.002$) reduction in risk was observed for the flavan-3-ol catechin when the highest and the lowest quintiles were compared. Conclusions: Anthocyanins and some flavone and flavan-3-ol compounds may contribute to the prevention of hypertension. These vasodilatory properties may result from specific structural similarities (including the B-ring hydroxylation and methyoxylation pattern).